

Bachelor Thesis

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Nomenclature

Footprint and Source Area Theory

η	Measured scalar concentration or flux
Ω_P	Source area of level P
Φ_{tot}	Total influence (integral of the source weight function)
f	Source weight function (footprint function)
L	Obukhov length (atmospheric stability parameter) [m]
P	Fractional contribution to total measured signal
Q	Source strength or emission rate

Physical and Flow Parameters

ν	Kinematic viscosity of air [m ² /s]
H	Reference building height [m]
k	Turbulent Kinetic Energy (TKE) [m ² /s ²]
Re	Reynolds number
u, v, w	Velocity components in x, y, z directions [m/s]
U_H	Mean wind velocity at building height [m/s]
$u_{rms}, v_{rms}, w_{rms}$	Root-mean-square velocity fluctuations [m/s]
z_0	Aerodynamic roughness length [m]

Statistical and Validation Metrics

D	Relative error tolerance (typically 0.25)
$FAC2$	Factor of two (fraction of data within a factor of two)
MAE	Mean Absolute Error
O_i	Observed value from AIJ experimental data
P_i	Predicted value from LBM simulation
q	Hit rate (fraction of predictions within tolerance)
$RMSE$	Root Mean Square Error
W	Absolute error tolerance (typically 0.05)

1 Introduction

1.1 Background

Air pollution in urban areas represents an ongoing global challenge that negatively impacts individual health, ecosystem integrity, and places increasing pressure on medical systems and economies worldwide. Urban air pollution comprises multiple components, including carbon dioxide (CO_2), carbon monoxide (CO), and particulate matter with an aerodynamic diameter of less than $2.5 \mu\text{m}$ ($\text{PM}_{2.5}$).

The health implications are severe; recent data from the World Health Organization (2024) indicates that ambient air pollution contributes to approximately 4.2 million premature deaths globally each year. While mortality is largely attributed to increased private vehicle demand and population growth, a significant contributing factor is the lack of consideration for pollutant dispersion during rapid urban development (Li et al., 2021). As noted by Li et al. (2021), "hasty and irrational construction strategies" often lead to poor ventilation, trapping pollutants in high concentrations near the ground level.

Understanding and mitigating these concentrations requires a dual-level approach. First, it involves the utilization of mechanical factors, which include forced convection by wind, natural convection by solar radiation, and traffic-induced turbulence. Second, it requires the strategic design of urban morphology. This focus on the "sound design of the urban landscape" emphasizes building geometries, urban density, and enclosure degrees to improve the city's natural capacity to flush out pollutants (Li et al., 2021).

The morphology and organization of urban areas directly influence urban traffic patterns and pollutant flow dynamics. Therefore, understanding the relationship between urban morphology, pollutant transport, and air quality is critical for developing effective mitigation strategies that move beyond qualitative observations toward quantitative air quality improvements.

1.2 Objective

2 Theory

2.1 Lattice Boltzmann Method

To simulate urban air pollutant transport, this study employs the Lattice Boltzmann Method (LBM), a mesoscopic computational fluid dynamics (CFD) technique. Unlike traditional solvers that directly discretize the macroscopic Navier-Stokes equations, LBM models fluid flow by simulating the collective behavior of a distribution of particles on a discrete grid, or "lattice" (Krüger et al., 2017). The movement of these particles is governed by the Lattice Boltzmann Equation (LBE), which describes the evolution of the discrete-velocity distribution function f_i :

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) - f_i(\mathbf{x}, t) = \Omega_i(f) + S_i \quad (1)$$

Where f_i is the distribution function representing the density of particles moving with velocity components $\mathbf{c}_i$ in the i -th direction. The term Ω_i represents the collision operator, which models the internal redistribution of particle momentum, while S_i is the source term. The inclusion of S_i is critical for this research as it allows for the integration of external body forces and the modeling of mass sources, such as vehicle emissions. The simulation proceeds through a sequence of two primary operations: streaming and collision. During the streaming step, particles move from their current node $\mathbf{x}$ to adjacent points based on their respective velocities. Following this, the collision step redistributes the particle velocities at each node based on local interactions, effectively mimicking fluid viscosity and relaxation toward equilibrium. As noted by Krüger et al. (2017), LBM offers several distinct advantages for simulating complex urban environments. First, the underlying Boltzmann equation is essentially a hyperbolic advection equation where the source terms depend only on local values of f . Because these operations rely strictly on information from a node and its immediate neighbors, the algorithm is inherently local. This locality facilitates high-performance parallelization, which is essential for large-scale urban simulations. Furthermore, the grid-based nature of LBM allows for the efficient handling of complex building geometries and irregular urban morphologies without the need for the intensive re-meshing often required by finite-element methods.

2.1.1 Lattice Configuration and Accuracy

The choice of the D3Q27 lattice model is fundamental to the accuracy of the turbulence modeling in this research (Yokouchi et al., 2020). While the simpler D3Q19 model is often used to reduce computational costs, it lacks the rotational invariance required to capture complex secondary flows accurately. As demonstrated by Suga et al. (2015), the D3Q27 lattice provides a superior representation of turbulent structures. Their comparative analysis of turbulent pipe flow showed that the D3Q27 model produces significantly more realistic flow fields than the D3Q19 model, which tends to struggle with resolving fine-scale turbulent fluctuations in three-dimensional space. By utilizing 27 discrete velocities, the model can more effectively resolve the secondary circulations and high-frequency fluctuations necessary for an in-depth understanding of the pollutant dispersion mechanism in urban areas. The internal redistribution of particle momentum is governed by the Bhatnagar-Gross-Krook (BGK) collision operator. This operator assumes that the particle populations relax toward a local equilibrium state, f_i^{eq} , at a rate controlled by a single relaxation time τ (Krüger et al., 2017):

$$\Omega_i(f) = -\frac{f_i - f_i^{eq}}{\tau} \Delta t \quad (2)$$

The equilibrium distribution f_i^{eq} is formulated as a second-order expansion of the Maxwell-Boltzmann distribution in terms of the macroscopic density ρ and velocity $\mathbf{u}$:

$$f_i^{eq}(\mathbf{x}, t) = w_i \rho \left(1 + \frac{\mathbf{u} \cdot \mathbf{c}_i}{c_s^2} + \frac{(\mathbf{u} \cdot \mathbf{c}_i)^2}{2c_s^4} - \frac{\mathbf{u} \cdot \mathbf{u}}{2c_s^2} \right) \quad (3)$$

Here, c_s represents the isothermal speed of sound, defined as $c_s^2 = \frac{1}{3} \frac{\Delta x^2}{\Delta t^2}$. This formulation ensures the local conservation of mass and momentum while establishing the relationship between pressure and density as $p = c_s^2 \rho$.

2.1.2 Large Eddy Simulation and Turbulence Modeling

To capture the transient, multi-scale nature of urban wind flows, this research utilizes a Large Eddy Simulation (LES) approach. The unresolved subgrid-scale (SGS) motions are accounted for using the Coherent-structure Smagorinsky Model (CSM). In a standard Smagorinsky framework, the turbulent viscosity ν_t is determined by the strain-rate tensor $|S|$ and a filter width Δ :

$$\nu_t = (C_S \Delta)^2 |S| \quad (4)$$

Within the LBM framework, the total viscosity ν (the sum of molecular and turbulent viscosity) is directly linked to the relaxation time:

$$\nu = c_s^2 \left(\tau - \frac{\Delta t}{2} \right) \quad (5)$$

While a standard Smagorinsky constant C_S typically ranges between 0.1 and 0.2 (Blazek, 2015), the CSM used here employs a dynamic parameter based on the second invariant of the velocity gradient tensor. This approach allows the model to automatically account for wall-damping effects, ensuring more accurate flow predictions near building surfaces without requiring manual damping functions (Phuc et al., 2018).

2.1.3 Implementation of Lagrangian Stochastic Model

To simulate the transport and dispersion of particulate matter within the urban environment, this study utilizes a Lagrangian Stochastic Model (LSM). Unlike Eulerian methods that treat pollutants as a continuous concentration field, the LSM tracks the trajectories of thousands of discrete "passive" particles within the turbulent flow field generated by the LBM. This approach is particularly effective for modeling dispersion in the inhomogeneous and non-Gaussian turbulence characteristic of the urban planetary boundary layer (PBL). By simulating a statistically significant ensemble of particle paths, the ensemble-mean concentration can be derived, which is directly proportional to the particle number density or the probability density function (PDF) of particle positions (Weil et al., 2004). The trajectory of each particle is determined by integrating its position x_p over time. For a particle starting at an initial position x_0 , the position at the next time step $t + \Delta t$ is calculated as:

$$x_p(x_0, t + \Delta t) = x_p(x_0, t) + u_p(x_0, t) \Delta t \quad (6)$$

where u_p represents the instantaneous Lagrangian particle velocity. This velocity is composed of two primary components:

$$u_p = u_r + u_s$$

In this formulation, u_r is the resolved grid-scale (GS) velocity obtained directly from the LBM-LES results, representing the large-scale atmospheric motions. The term u_s represents the subgrid-scale (SGS) velocity fluctuations, which account for the smaller, unresolved turbulent motions. The resolved velocity u_r at the particle's location is determined by interpolating the LBM grid results using the weighting factors w_i of the neighboring nodes:

$$u_r = \sum_{i=0}^7 (w_i U_i)$$

To account for the unresolved fluctuations, the SGS velocity u_s is modeled as a stochastic process. Following the framework established by Weil et al. (2004) and Thomson (1987), the evolution of the SGS velocity du_s is described by a stochastic differential equation:

$$du_s = -\frac{3f_s C_0 \epsilon}{4} dt + \frac{1}{2} \left(\frac{1}{e_s} \frac{de_s}{dt} u_s + \frac{2}{3} \nabla e_s \right) dt + (f_s C_0 \epsilon)^{1/2} d\xi \quad (7)$$

where e_s is the local SGS turbulent kinetic energy (TKE) and f_s represents the mean contribution of the SGS TKE to the total TKE. The term ϵ denotes the total dissipation rate, while C_0 is a universal constant (set to 4.0 in this study). The stochastic component $(f_s C_0 \epsilon)^{1/2} d\xi$ incorporates Gaussian white noise via the displacement vector $d\xi$, allowing the model to represent the random nature of sub-lattice turbulent dispersion.

2.1.4 Advanced Wall Boundary Conditions

In LBM, the standard Bounce-Back (BB) scheme assumes that fluid particles hitting a solid boundary reverse their momentum, effectively enforcing a no-slip condition ($\mathbf{u} = 0$) at the wall. While numerically stable and simple to implement, the BB scheme requires the first grid point to reside within the viscous sublayer to accurately capture wall shear stress. In urban-scale simulations, the grid resolution is typically much larger than the thickness of this sublayer, causing the BB scheme to underpredict shear drag. To resolve this, a Wall Function Boundary (WFB) is implemented, following the approach evaluated by Han et al. (2021). The WFB utilizes Spalding's Law to model the non-linear velocity profile in the logarithmic layer, allowing the simulation to approximate near-wall velocity without fully resolving the thin viscous region. This modification is applied during the collision step by introducing a momentum loss term based on the wall shear stress τ_w :

$$f_a^* = f_a \pm \frac{\Delta t}{2\Delta y} \tau_{w,j} \quad (8)$$

where $\tau_{w,j}$ is the shear stress in the j -direction (x or z). The wall shear stress is determined by solving Spalding's implicit equation:

$$y^+ = u^+ + e^{-\kappa B} \left[e^{\kappa u^+} - 1 - (\kappa u^+) - \frac{(\kappa u^+)^2}{2} - \frac{(\kappa u^+)^3}{6} \right] \quad (9)$$

Here, y^+ is the dimensionless distance from the wall, and u^+ is the dimensionless velocity. By calculating the friction velocity $u_\tau = \sqrt{\tau_w / \rho}$ from this relationship, the model applies a "reverse resistance" that decelerates particles near building surfaces. As demonstrated by Han et al. (2021), this WFB approach significantly improves the accuracy of shear drag predictions in high-Reynolds-number urban flows where coarse grids are a necessity.

2.1.5 Subgrid-Scale Turbulent Kinetic Energy (SGS TKE)

The calculation of the SGS TKE (e_s) is vital for the Lagrangian Stochastic Model, as it dictates the magnitude of the stochastic velocity fluctuations. While traditional LES often parameterizes e_s via the eddy viscosity and a mixing length ($v_t = c_k l e_s^{1/2}$), this study utilizes a filtering approach as implemented by Suga et al. (2015) and Yokouchi et al. (2020). The SGS TKE (k_{sgs}) is derived from the difference between the instantaneous grid-scale velocity and a filtered velocity field:

$$k_{sgs} = C_{kes} \sum_{i=1}^3 (\langle \hat{U}_i \rangle - \langle U_i \rangle)^2 \quad (10)$$

The filtered velocity $\langle \hat{U}_i \rangle$ is calculated using a weighted average of the central node (50% weight) and its six immediate neighbors (1/12 weight each):

$$\langle \hat{U}_i \rangle = \frac{1}{2} \langle U_i \rangle + \frac{\langle U_i^E \rangle + \langle U_i^W \rangle + \langle U_i^N \rangle + \langle U_i^S \rangle + \langle U_i^T \rangle + \langle U_i^B \rangle}{12} \quad (11)$$

This local filtering operation is highly compatible with the LBM's neighbor-only communication pattern, maintaining the algorithm's high parallel efficiency while providing a robust estimate of local turbulence intensity.

2.1.6 Stochastic Integration and Stability

The stochastic component of the LSM involves the integration of Gaussian white noise, $\xi(t)$, which is formally defined as the time derivative of a Wiener process $W(t)$:

$$W_i(t) = \int_0^t \xi(s) ds \quad (12)$$

In accordance with Brownian motion theory, the variance of this process increases linearly with time. Given that the SGS velocity u_s is modeled as a random walk, the stochastic term $(f_s C_0 \epsilon)^{1/2} d\xi$ can occasionally produce unrealistically high instantaneous velocities. This is particularly prevalent near building walls where e_s is high and the gradient terms in the LSM equation are steep. To prevent numerical instability and maintain physical consistency, a velocity limitation is enforced, ensuring that the total Lagrangian particle velocity u_p remains within physically plausible bounds relative to the local mean flow Yokouchi et al., 2020.

2.1.7 Model Selection and Validation via Flat Plate Analysis

Before applying the model to complex urban morphologies, Yokouchi et al. (2020) conducted a validation study using particle dispersion over a flat plate to identify the most accurate combination of numerical schemes. The evaluation compared three primary configurations: (1) a baseline model using a simple Bounce-Back (BB) boundary and the standard subgrid-scale (SGS) model; (2) a model incorporating the Wall Function (WF) and Suga's refined SGS TKE calculation; and (3) an integrated model combining the WF-Suga framework with the Lagrangian Stochastic Model (LSM) and a velocity limitation. The accuracy of these models was evaluated based on their ability to replicate the theoretical logarithmic profile of particle concentration in a neutral boundary layer:

$$\frac{C_0 - C}{C^*} = \frac{1}{\kappa} \ln \frac{z}{z_0} \quad (13)$$

where C_0 represents the particle density at ground level, C^* is the friction density, κ is the von Kármán constant, and z_0 is the roughness length. The results demonstrated that the WF-Suga-LSM model provided the highest correlation with the logarithmic profile. In contrast, models without the LSM or the specialized SGS TKE tended to deviate from the expected distribution, particularly in the near-wall region. Furthermore, the necessity of the speed limitation—defined as $|u_s| < k_{sgs}^{1/2}$ —was confirmed through an analysis of the particle velocity variance. Without this limitation, the stochastic term in the LSM produced unrealistically high velocity fluctuations near the boundary. Implementing the speed limit successfully reduced this variance, ensuring a stable and physically consistent velocity distribution across the vertical profile. By adopting this validated WF-Suga-LSM framework with speed limitations, this research ensures that the fundamental transport mechanisms are correctly modeled before introducing the complexities of the AIJ Case H urban block.

2.2 Source Area Modeling for Scalars and Fluxes

In micrometeorological measurements, particularly within complex urban environments, it is critical to determine the spatial representativeness of data collected at a specific height. While traditional models often assume homogeneous surfaces, real-world applications require an understanding of which specific surface areas influence a sensor. Schmid (1994) addressed this by defining the Source Area as the surface region contributing to a measured scalar concentration or flux, using the concept of a Source Weight Function (SWF).

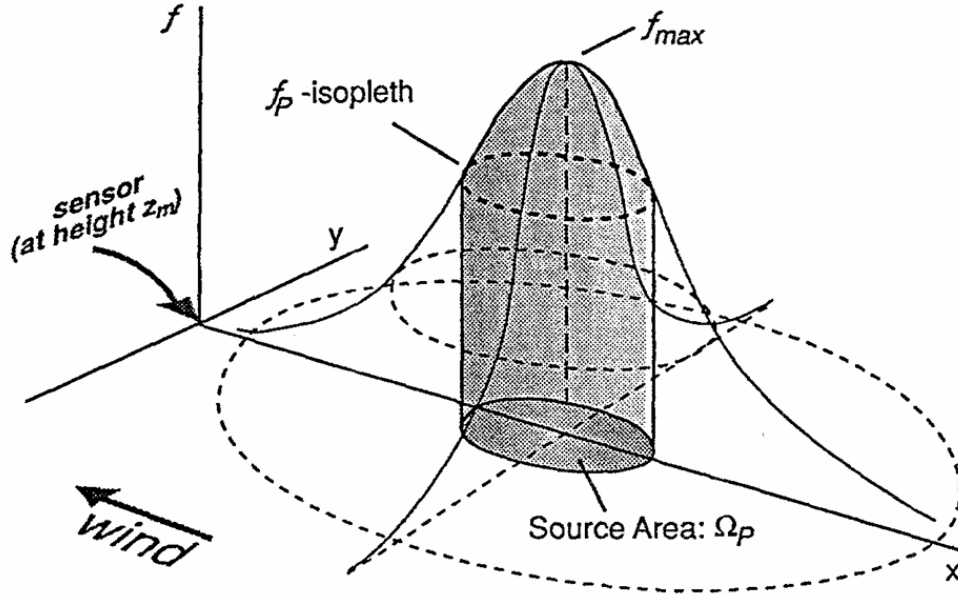


Figure 1: Conceptual illustration of the source area model, adapted from Schmid (1994). The 3D surface represents the source weight function, which shows how much each ground location influences the sensor. The influence is small near the sensor, reaches a peak at a certain upwind distance, and then falls off. The shaded region under the horizontal slice (isopleth f_P) represents the fraction P of the total influence. The source area Ω_P is the two-dimensional region on the x - y plane formed by the downward projection of this slice.

The relationship between a measurement at a specific point and the surface sources is fundamentally

described by a convolution integral. Let $\eta(\mathbf{r})$ represent a measured quantity at position $\mathbf{r}$, and $Q(\mathbf{r}')$ represent the source strength at position $\mathbf{r}'$. The measurement is linked to the source via a probability transfer function, or Green's function, f :

$$\eta(\mathbf{r}) = \int_R Q(\mathbf{r}') f(\mathbf{r} - \mathbf{r}') d\mathbf{r}' \quad (14)$$

In this context, f represents the probability that material emitted at the surface location $\mathbf{r}'$ will arrive at the sensor location $\mathbf{r}$.

For measurements taken in the atmospheric surface layer at a height z_m , the general 3D convolution is reduced to a 2D surface integral. Here, z_0 represents the surface roughness length:

$$\eta(0, 0, z_m) = \int_{-\infty}^{\infty} \int_{-\infty}^x Q_\eta(x, y, z_0) f(-x, -y, z_m - z_0) dx dy \quad (15)$$

To represent a specific point source at coordinates (x_s, y_s, z_0) with strength Q_u , the source term is defined using Dirac delta distributions (δ):

$$Q(x, y, z_0) = Q_u \delta(x - x_s) \delta(y - y_s) \delta(z - z_0) \quad (16)$$

By substituting the point source definition into the integral, the relationship simplifies to a direct link between the source strength and the sensor through the source weight function:

$$\eta(0, 0, z_m) = Q_u f(x_s, y_s, z_m - z_0) \quad (17)$$

The "Source Area" is defined as the smallest surface area that accounts for a specified fraction P (e.g., 50%) of the total influence on the measured value. This is calculated by integrating the source weight function over the region where its values exceed a threshold f_P :

$$P = \frac{1}{\Phi_{tot}} \iint_{f(x,y) \geq f_P} f(x, y) dx dy \quad (18)$$

where the total influence Φ_{tot} is the integral of $f(x, y)$ over the entire surface:

$$\Phi_{tot} = \iint f(x, y) dx dy \quad (19)$$

3 Methodology

3.1 Validation and Numerical Workflow

3.1.1 Benchmark Case: AIJ Case H

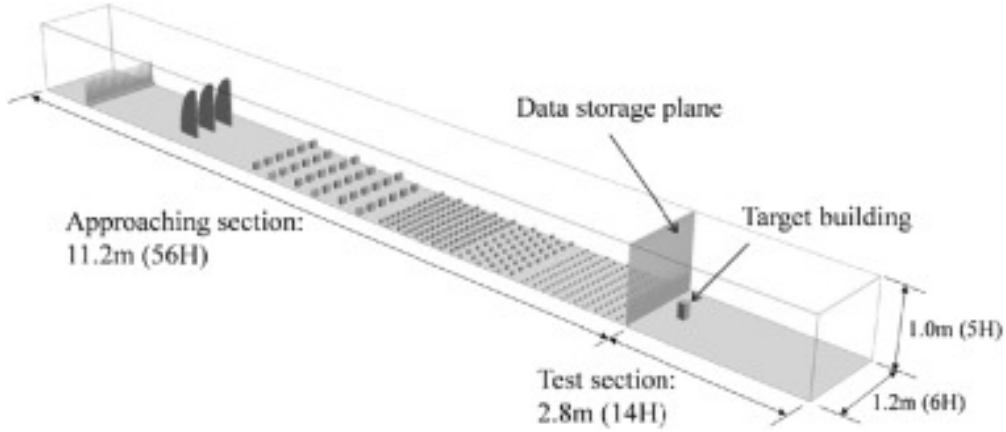


Figure 2: Experimental setup for AIJ Case H at Tokyo Polytechnic University, isolated building model and the arrangement of spires and roughness elements in the approach section to induce a turbulent boundary layer (adapted from Architectural Institute of Japan, 2021; Okaze et al., 2021).

The numerical validation of the Lattice Boltzmann Method (LBM) in this study is based on Case H of the Architectural Institute of Japan (AIJ) benchmark, which involves an isolated building model within a turbulent boundary layer. The physical experiment was conducted at Tokyo Polytechnic University using a 1:250 scale model. The building is a rectangular prism with a height-to-width ratio of 1 : 1 : 2 ($H = 0.20$ m). The wind tunnel test section measures $2.8\text{ m} \times 1.2\text{ m} \times 1.0\text{ m}$, corresponding to normalized dimensions of $14H \times 6H \times 5H$. Turbulence in the approach section was induced using spires and roughness elements to simulate a realistic atmospheric boundary layer. For the pollutant dispersion study, a tracer gas release point was located 50 mm ($0.25H$) from the leeward side of the building. To ensure comparability across different scales, the reference velocity U_H is defined as the mean inflow velocity at the building height.

The LBM results in this research are cross-validated against the comprehensive Large Eddy Simulation (LES) study conducted by Okaze et al. (2021). Their study utilized OpenFOAM to investigate the influence of various subgrid-scale (SGS) models—including the Standard Smagorinsky (S), Dynamic Smagorinsky (DS), Wall-Adapting Local Eddy-Viscosity (WALE), and Coherent Structure (CS) models—on turbulent statistics. Notably, Okaze et al. (2021) evaluated grid sensitivity across 10, 20, and 40-grid resolutions relative to the building width. This thesis focuses on comparing LBM performance against the validation study, using the same validation standards defined in the paper, which is a hit rate (q) of ≥ 0.66 and a factor of two observations (FAC2) of > 0.5 . By comparing the LBM-LSM results directly to these established OpenFOAM benchmarks, the predictive accuracy of the model can be evaluated.

A significant factor in the accuracy of the benchmark study was the choice of the computational grid. Okaze et al. (2021) utilized a domain of $12.5H(x) \times 6.0H(y) \times 5.0H(z)$ and intentionally avoided non-conformal grids—setups where the grid is made finer or coarser in specific areas to save on calculation time. They observed that sudden changes in grid size can cause mathematical errors and oscillations (unstable values) at the boundaries where the different grid scales meet. To ensure their comparisons of the physics

models remained pure, they used a consistent, orthogonal grid system throughout their study.

In this research, the LBM-LES solver employs a uniform grid that remains perfectly even across the entire simulation space. This grid is technically non-conformal to the building's geometry, but do not bend or wrap to match the building's surface so it avoids the errors mentioned by Okaze et al. (2021). Because every grid cell in the LBM simulation is the same size regardless of its location or proximity to the building, the wind field calculation maintains a high degree of mathematical stability.

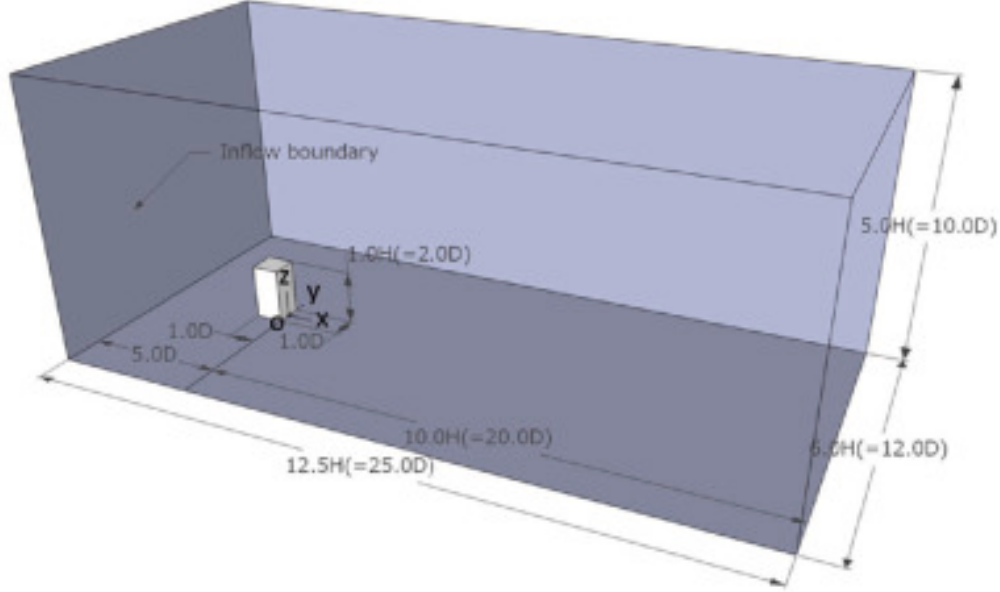


Figure 3: Numerical domain configuration and measurement locations for the LES validation study (adapted from Okaze et al., 2021).

3.1.2 Computational Implementation and Workflow

The simulation is designed to run efficiently by using the Graphics Processing Units (GPU) to handle the heavy mathematical workload. The source code is written using the CUDA platform, which allows a standard programming language like C to run thousands of calculations simultaneously on the GPU hardware (NVIDIA Corporation, 2024). The software is organized into modules where the wind simulation (LBM) and the particle tracking (LPT) are connected. The wind field is calculated first, and that information is then used to move the particles. The system is managed by configuration files (ending in `.mak`) and is organized into six functional parts:

1. **Simulation Setup and Constants:** This part handles the initial settings, such as the size of the wind domain and the physical properties of the air and particles. It uses header files (like `Define_user.h`) to set the desired variables for the specific research case before the calculation begins.
2. **Data Memory Management:** Before calculations start, information must be moved from the computer's main memory to the GPU's high-speed memory. This module ensures that the wind grid and particle positions are stored where the GPU can access them quickly.
3. **Wind Field Calculation (LBM):** This is the core of the wind solver. It calculates how the air moves and collide on a grid. Because these collisions happen at every point in space at the same time, the

GPU handles this by calculating every grid cell in parallel.

4. **Boundary and Obstacle Handling:** This module manages how the wind interacts with boundaries, such as the ground surface or the building model. It applies the specific roughness settings (z_0) and ensures the wind cannot pass through solid walls.
5. **Particle Movement (LPT):** Once the wind speed is known at a specific point, this module calculates the new position of each individual particle. It accounts for the forces the wind exerts on the particles, moving them through the complex flow patterns around the building.
6. **Data Output and Recording:** At set intervals, the simulation saves the state of the wind and particles to the storage system. These are the files (such as the .csv and binary results) used for the statistical validation and profile comparisons discussed in the results section.

The graphical overview of the system architecture described above can be seen in Figure 4.

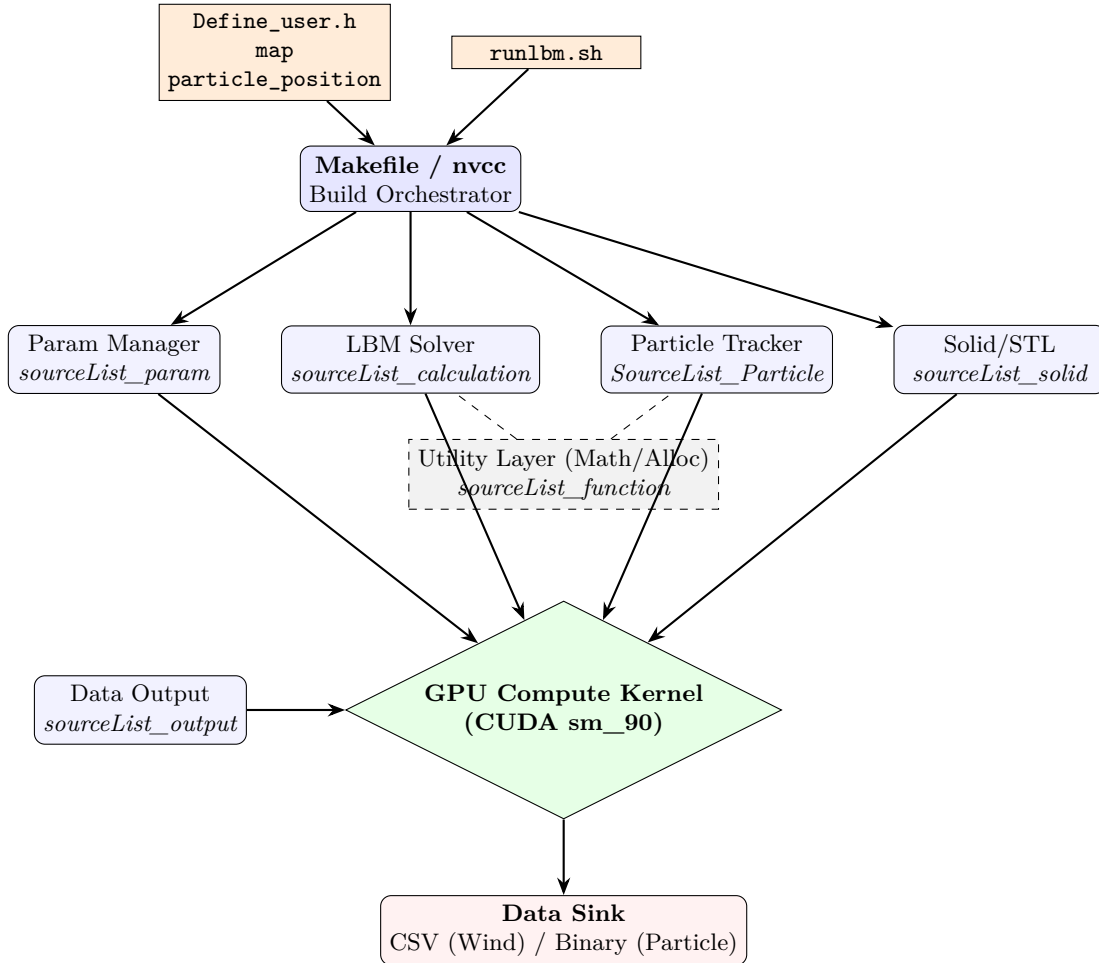


Figure 4: System architecture of the LBM simulation framework

The same source code for the simulation can also be used when running the simulation on a larger scale. However, to handle an even larger calculation and longer periods of time, multiple nodes and GPUs must be used. As such, even though the base code and the output process is the same, the code is ran in the

TSUBAME 4.0 Supercomputer. A shell script `run_program.sh` is used to queue the LBM-LSM simulation program into the supercomputer job scheduler, before the simulation is run. The system architecture diagram of this process can be viewed in Figure 5.

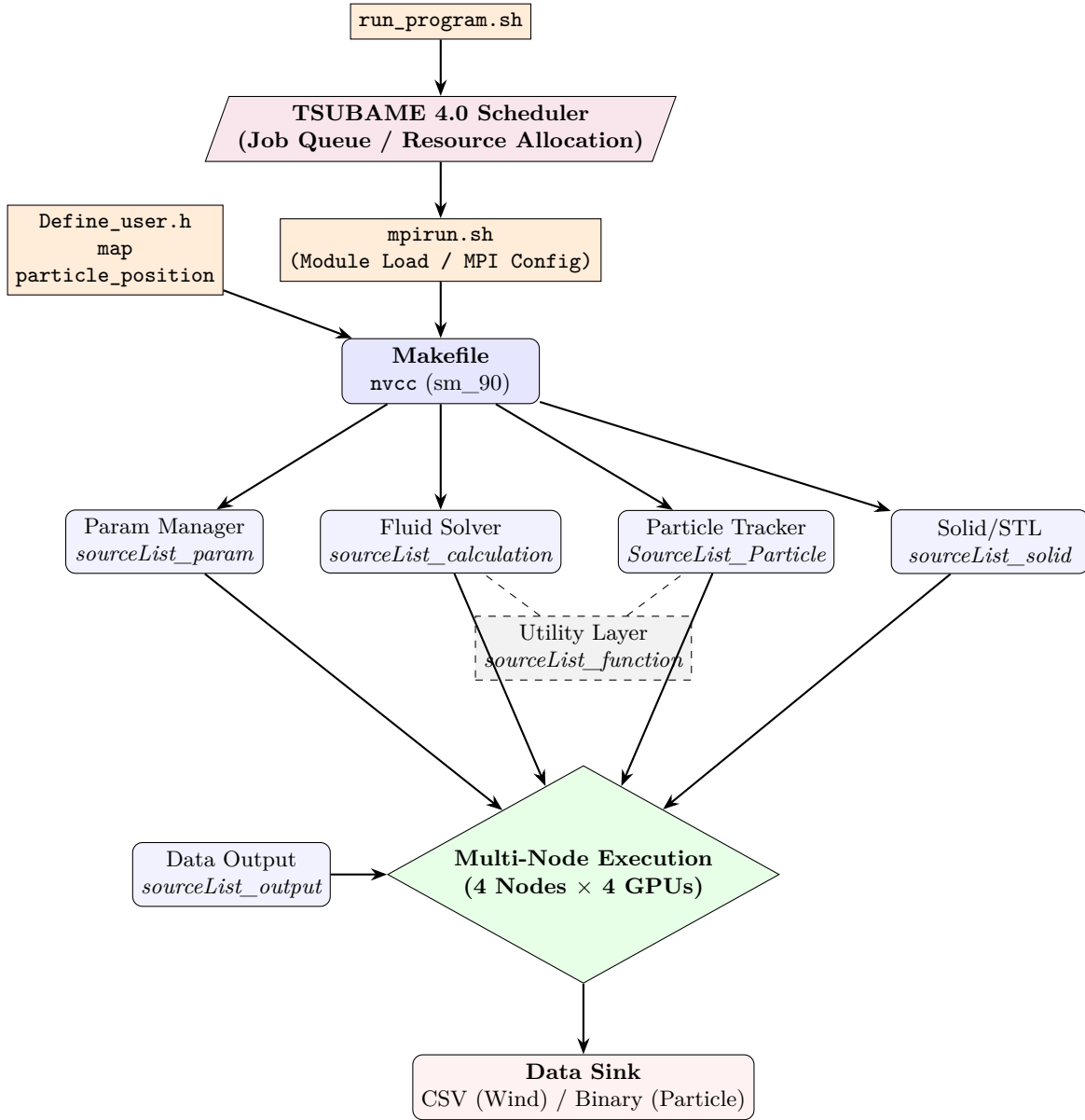


Figure 5: System architecture of the LBM simulation framework executed using the TSUBAME 4.0 Supercomputer

3.2 Simulation Parameters and Case Setup

The specific configurations for the validation and research cases are summarized in Table 1. These cases represent the transition from local prototyping to large-scale supercomputer execution.

Table 1: Detailed Simulation Case Parameters

Case	Environment	Domain Size (x, y, z in m)	Duration (s)	Approach	H_{cube} (m)	z_0 (m)	U_0 (m/s)	Flow Rate
1	Local Server	$832 \times 384 \times 240$	2400	No	—	0.001	6.0	150 p/s
2	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	8	0.001	6.0	150 p/s
3	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	16	0.001	6.0	150 p/s
4	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	20	0.001	6.0	150 p/s
5	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	24	0.001	6.0	150 p/s
6	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	32	0.001	6.0	150 p/s
7	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	16	0.1	6.0	150 p/s
8	TSUBAME 4.0	$4352 \times 384 \times 320$	6000	Yes	20	0.1	6.0	150 p/s

3.3 Physical and Dimensionless Parameters

To ensure the dynamic similarity between the wind tunnel experiment and the Lattice Boltzmann Method (LBM) simulation, the Reynolds number (Re) is calculated. This dimensionless value represents the ratio of inertial forces to viscous forces.

Reynolds Number (Re) For this study, the characteristic length is defined as the building height H , and the velocity is the wind speed at that height U_H :

$$Re = \frac{U_H \cdot H}{\nu} \quad (20)$$

While the experimental Re_{exp} is 4.3×10^4 , the LBM simulation on TSUBAME 4.0 operates at a significantly higher effective Reynolds number ($Re_{LBM} \approx 2.09 \times 10^7$). This discrepancy is strategically justified by the physical principles of urban flow modeling. According to the guidelines established by Snyder (1981), a Reynolds number of 1.1×10^4 serves as a conservative threshold for ensuring that the flow characteristics around cube-like models are independent of the specific Re value.

Since both the experiment and the simulation significantly exceed this critical threshold, the large-scale turbulent structures that serve as the primary focus of the Large Eddy Simulation (LES) approach remain physically consistent. The higher Re_{LBM} utilized in this thesis reflects the high-resolution grid permitted by the supercomputing environment, allowing for a more detailed capture of the flow without violating the principle of Reynolds-number independence.

3.3.1 Wind and Turbulence Data Post-Processing and Statistical Analysis

The Large Eddy Simulation (LES) results were post-processed to extract time-averaged wind statistics and turbulence characteristics. The simulation domain was decomposed into four parallel sub-domains (indexed 0000 to 0003) for computational efficiency on the TSUBAME 4.0 supercomputer.

Data analysis focused primarily on the fourth sub-domain (0003), which contains the building model and the validation points of interest, excluding the approach flow sections (0000-0002).

Data Structure and Extraction Wind field data were exported as planar cross-sections (xy -planes) at various vertical grid heights (k). The raw output files follow a stacked structure where each height level is preceded by a header indicating the vertical grid index, followed by a 192×544 matrix representing the spanwise (y) and streamwise (x) velocity components. On the small domain simulation (Case 1), the data structure is similar, except for the inexistence of the approach section. Thus, for the small domain simulation, only one sub-domain, 0000, exists, and the analysis is done over that sub-domain only.

The analysis utilizes time-averaged velocity components ($\bar{u}, \bar{v}, \bar{w}$) and second-order velocity moments ($\overline{u^2}, \overline{v^2}, \overline{w^2}$), averaged over 540,000 time steps (equivalent to 5,400 physical seconds) to ensure statistical

convergence. These variables correspond to the output files `xy_um`, `xy_vm`, `xy_wm` (first-order moments) and `xy_uu`, `xy_vv`, `xy_ww` (second-order moments).

Turbulence Statistics To evaluate the turbulent flow characteristics, Reynolds stresses were calculated by subtracting the square of the mean velocity from the mean of the squared velocity. The streamwise ($\langle u'^2 \rangle$), spanwise ($\langle v'^2 \rangle$), and vertical ($\langle w'^2 \rangle$) Reynolds normal stresses were derived as follows:

$$\langle u'^2 \rangle = \overline{u^2} - (\bar{u})^2 \quad (21)$$

$$\langle v'^2 \rangle = \overline{v^2} - (\bar{v})^2 \quad (22)$$

$$\langle w'^2 \rangle = \overline{w^2} - (\bar{w})^2 \quad (23)$$

Root Mean Square (RMS) Fluctuations The Root Mean Square (RMS) velocity fluctuations, which indicate the intensity of turbulence in each direction, were calculated and normalized by the reference approach flow velocity (U_H) at building height H :

$$u_{rms} = \frac{\sqrt{\langle u'^2 \rangle}}{U_H}, \quad v_{rms} = \frac{\sqrt{\langle v'^2 \rangle}}{U_H}, \quad w_{rms} = \frac{\sqrt{\langle w'^2 \rangle}}{U_H} \quad (24)$$

Turbulence Kinetic Energy (TKE) The Turbulence Kinetic Energy (k), representing the mean kinetic energy per unit mass associated with eddies in the turbulent flow, was computed as half the sum of the Reynolds normal stresses:

$$k = \frac{1}{2} (\langle u'^2 \rangle + \langle v'^2 \rangle + \langle w'^2 \rangle) \quad (25)$$

To handle numerical stability during post-processing, a constraint was applied such that any negative stress values resulting from floating-point errors were clamped to zero before the square root operation ($\max(0, \sigma^2)$). All statistical results were mapped to the physical domain coordinates to facilitate direct comparison with the Architectural Institute of Japan (AIJ) wind tunnel experimental benchmarks.

3.3.2 Lagrangian Particle Data Post-Processing and Statistical Analysis

The raw simulation output is generated in a condensed binary format to manage the high volume of data produced by the Lagrangian tracking. These binary files are recorded at 1-second intervals (equivalent to every 100 simulation time steps). For each output interval, a set of specialized files is created, including `header`, `index`, `position`, `velocity`, and `SGS_velocity`. Each file contains the specific physical attributes of all particles active within the domain at that specific moment. For instance, a file named `position0-001.bin` stores the three-dimensional Cartesian coordinates (x, y, z) of every particle at $t = 1$ second.

To translate these large-scale datasets into physically interpretable results, a dedicated analysis suite was developed in C++. This suite utilizes a modular design where specific physical quantities are calculated by independent components managed by a central execution program. The workflow is handled by a C++ executable, compiled using a `Makefile` that links distinct modules for density calculation, flux analysis, vertical profiling, and Flux Footprint Prediction (FFP).

The system is designed for high flexibility across different research scenarios. Rather than fixed internal parameters, the tool employs a command-line interface where the domain dimensions, grid resolution, time steps, and output intervals are provided as arguments at the time of execution. This allows the same

analytical core to be applied to various cases—ranging from local isolated building tests to large-scale urban simulations—without requiring the code to be re-compiled.

The analysis is typically managed through batch scripts (`run.bat`), allowing the framework to simultaneously perform multiple tasks, such as extracting particle density profiles at specified vertical heights (z_{out}) and computing source area contributions for defined sensor locations.

To process the millions of Lagrangian trajectories efficiently, the post-processor utilizes OpenMP for multi-threading (`OMP_NUM_THREADS`). This allows the software to use multiple CPU cores simultaneously to read the binary data and perform calculations in parallel. The final results are aggregated into averaged statistical summaries in `.csv` or text format, which are then used for final visualization and plotting in Python.

Normalized Concentration (C_{norm})

To allow for a direct comparison between experimental gas data and the Lagrangian particle model, concentration values are normalized based on the flow rate Q , velocity U_H , and building scale H .

$$C_{norm} = \frac{C_o \cdot U_H \cdot H^2}{Q} \quad (26)$$

3.4 Statistical Validation Metrics

Following the AIJ guidelines for urban wind environments, four metrics are used to compare the predicted values (P_i) against the observed experimental data (O_i).

Hit Rate (q) The Hit Rate measures the reliability of the model. A data point is a "hit" if the prediction is within a relative tolerance D (usually 0.25) or an absolute tolerance W (usually 0.05).

$$q = \frac{1}{N} \sum_{i=1}^N n_i, \quad n_i = \begin{cases} 1 & \text{if } \left| \frac{P_i - O_i}{O_i} \right| \leq D \text{ or } |P_i - O_i| \leq W \\ 0 & \text{else} \end{cases} \quad (27)$$

Factor of Two (FAC2) FAC2 is a robust measure of the overall spread. It checks if the predicted value is between half and double the observed value.

$$\text{FAC2} = \frac{1}{N} \sum_{i=1}^N n_i, \quad n_i = \begin{cases} 1 & \text{if } 0.5 \leq \frac{P_i}{O_i} \leq 2.0 \\ 0 & \text{else} \end{cases} \quad (28)$$

Error Metrics (MAE and RMSE) The Mean Absolute Error (MAE) provides the average magnitude of the errors, while the Root Mean Square Error (RMSE) gives a higher weight to large outliers, providing a stricter assessment of simulation accuracy.

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |P_i - O_i|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (P_i - O_i)^2} \quad (29)$$

4 Results and Discussion

4.0.1 Validation of Wind and Turbulence Statistics

Table 2: Full validation results (q and $FAC2$) for velocity components and turbulence metrics across all approach flow cases. Underlined values indicate performance below the standard thresholds ($q < 0.66$, $FAC2 < 0.50$).

Case	u		v		w		u_{rms}		v_{rms}		w_{rms}		TKE	
	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$	q	$FAC2$
Case 1: No Approach	<u>0.533</u>	0.700	0.933	0.978	<u>0.389</u>	<u>0.433</u>	<u>0.622</u>	0.622	<u>0.644</u>	0.644	<u>0.611</u>	0.667	<u>0.489</u>	0.589
Case 2: 8m roughness	0.900	0.956	0.933	0.989	<u>0.456</u>	0.589	0.944	0.944	1.000	1.000	0.956	1.000	<u>0.611</u>	0.678
Case 3: 16m roughness	0.911	0.967	0.978	1.000	<u>0.533</u>	0.567	1.000	1.000	1.000	1.000	0.956	1.000	0.933	1.000
Case 4: 20m roughness	<u>0.822</u>	0.933	0.900	0.967	<u>0.567</u>	0.600	1.000	1.000	1.000	1.000	1.000	1.000	0.978	1.000
Case 5: 24m roughness	0.900	0.978	0.922	0.967	0.711	0.756	1.000	1.000	0.967	1.000	0.900	1.000	<u>0.656</u>	0.978
Case 6: 32m roughness	0.933	0.978	0.822	0.978	<u>0.578</u>	0.633	0.978	1.000	0.867	1.000	0.856	1.000	<u>0.478</u>	0.833
Case 7: 16m roughness - 0.1 z0	<u>0.867</u>	0.922	0.878	0.978	0.689	0.678	1.000	1.000	1.000	1.000	0.967	1.000	0.922	0.933
Case 8: 20m roughness - 0.1 z0	0.911	0.956	0.967	1.000	0.756	0.644	1.000	1.000	1.000	1.000	0.922	1.000	0.811	1.000

The validation results highlight the critical role of approach flow modeling in the validation study. While Case 1 (No Approach) consistently fails the 0.66 hit-rate (q) threshold across several components, the introduction of surface roughness in Cases 2-8 brings the streamwise velocity (u) and fluctuation components (u_{rms} , v_{rms} , w_{rms}) into strong alignment with experimental standards. Notably, vertical velocity (w) and Turbulent Kinetic Energy (TKE) remain the most sensitive parameters, occasionally falling below the standard in lower-roughness scenarios, but showing marked improvement as roughness height increases. This suggests that the LBM solver combined with the TSUBAME 4.0 is highly capable of capturing complex urban wind fields, provided the inflow turbulence is correctly parameterized.

Table 3: MAE and RMSE validation metrics for velocity components and TKE across all approach flow cases.

Case	u		v		w		u_{rms}		v_{rms}		w_{rms}		TKE	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Case 1: No Approach	0.2009	0.2726	0.0202	0.0267	0.0979	0.1212	0.0759	0.0996	0.0585	0.0675	0.0539	0.0619	0.0268	0.0320
Case 2: 8m roughness	0.0751	0.1518	0.0171	0.0232	0.0585	0.0705	0.0438	0.0575	0.0433	0.0485	0.0394	0.0450	0.0220	0.0254
Case 3: 16m roughness	0.0600	0.1286	0.0133	0.0184	0.0548	0.0650	0.0243	0.0304	0.0354	0.0407	0.0289	0.0341	0.0172	0.0208
Case 4: 20m roughness	0.0690	0.1266	0.0271	0.0321	0.0461	0.0538	0.0232	0.0296	0.0218	0.0259	0.0261	0.0307	0.0123	0.0150
Case 5: 24m roughness	0.0698	0.1194	0.0248	0.0304	0.0354	0.0429	0.0289	0.0366	0.0466	0.0518	0.0408	0.0471	0.0236	0.0286
Case 6: 32m roughness	0.0527	0.0925	0.0290	0.0338	0.0419	0.0501	0.0447	0.0537	0.0470	0.0537	0.0463	0.0524	0.0294	0.0348
Case 7: 16m roughness - 0.1 z0	0.0843	0.1398	0.0247	0.0317	0.0387	0.0459	0.0304	0.0401	0.0215	0.0251	0.0310	0.0366	0.0135	0.0161
Case 8: 20m roughness - 0.1 z0	0.0714	0.1238	0.0145	0.0184	0.0364	0.0415	0.0241	0.0291	0.0256	0.0328	0.0269	0.0377	0.0149	0.0206

The quantitative assessment of the error metrics further supports the conclusions drawn from the hit rate analysis. In Case 1, the RMSE for streamwise velocity u is notably high at 0.2726, representing a significant deviation from experimental values. However, across the roughness-modified Cases 2-8, the RMSE for u is reduced by over 50% in most instances, reaching its minimum in Case 6 (0.0925). Interestingly, the vertical velocity components (v , w) and turbulence fluctuations show relatively stable and low MAE values (< 0.05) across the majority of the cases. The TKE errors are particularly low in the optimized roughness scenarios (e.g., Case 4 with an RMSE of 0.0150), demonstrating that the solver’s ability to capture the energy distribution within the flow field is highly dependent on the correct specification of the approach flow boundary layer. These low absolute errors confirm that the model’s predictions are not only statistically correlated but also physically accurate in magnitude.

The concentration validation results show a marked contrast to the velocity validation, with significantly lower accuracy across all metrics. For all simulation cases, the hit rate (q) remains well below the 0.66

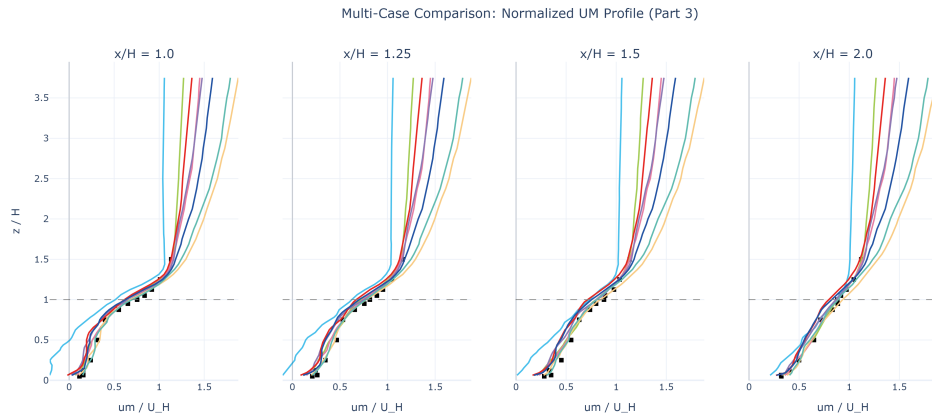
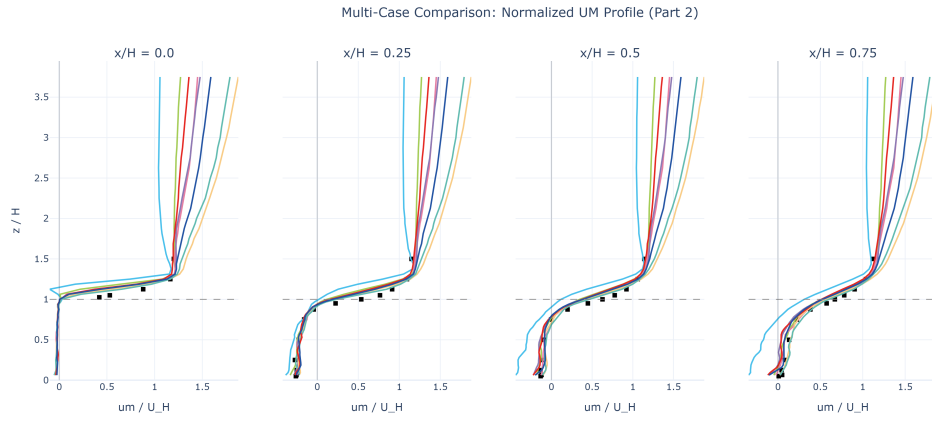
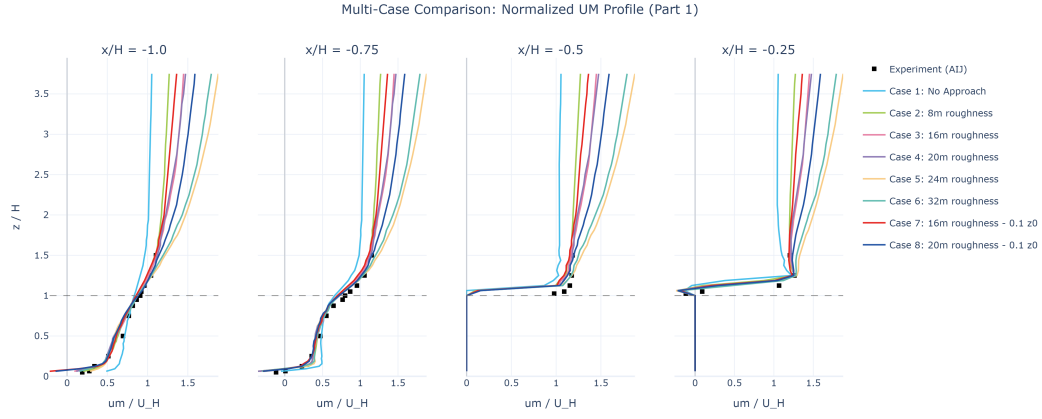
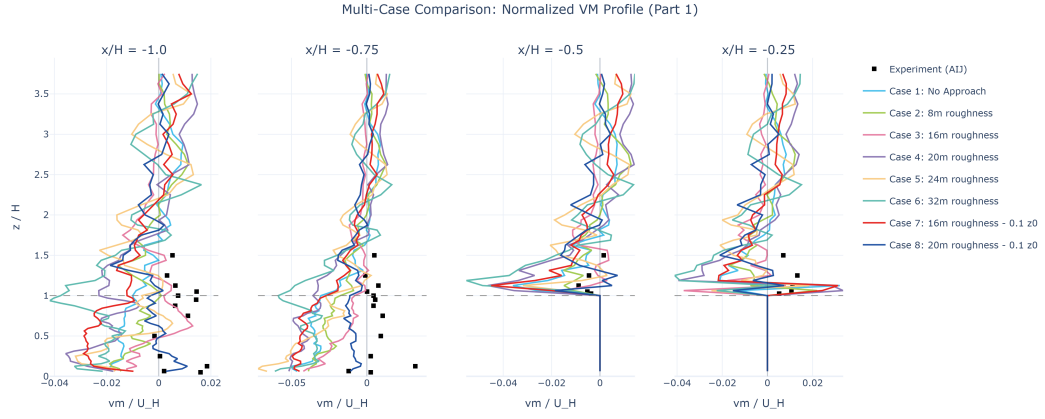
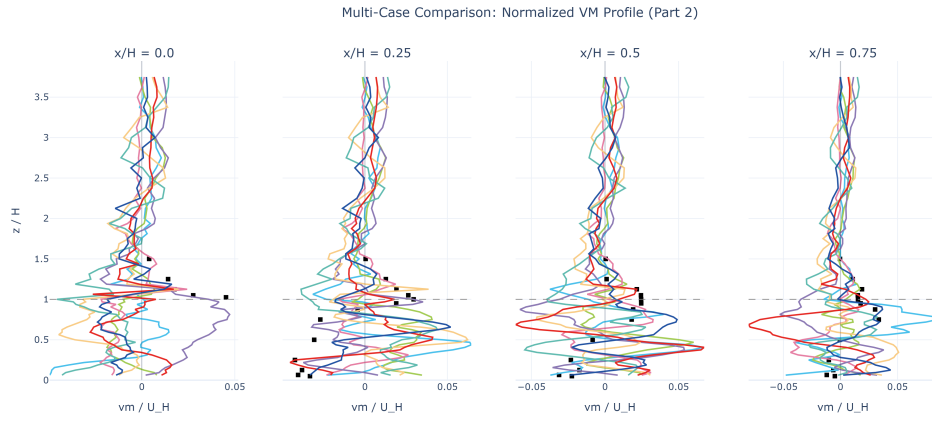


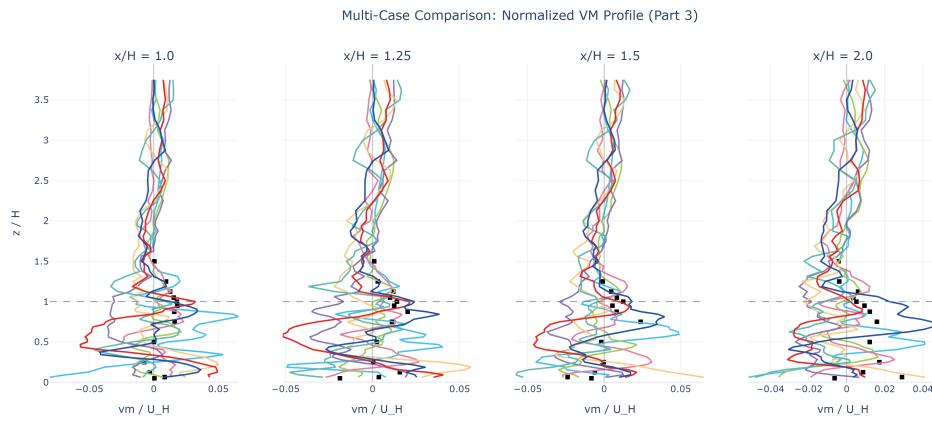
Figure 6: Normalized mean streamwise velocity ($\bar{u}/U_H$) profiles at (a) Windward locations ($x/H = -1.0$ to -0.25), (b) Near Wake locations ($x/H = 0.0$ to 0.75), and (c) Far Wake locations ($x/H = 1.0$ to 2.0).



(a)



(b)



(c)

Figure 7: Normalized mean spanwise velocity ($\bar{v}/U_H$) profiles at (a) $x/H = -1.0$ to -0.25 , (b) $x/H = 0.0$ to 0.75 , and (c) $x/H = 1.0$ to 2.0 .

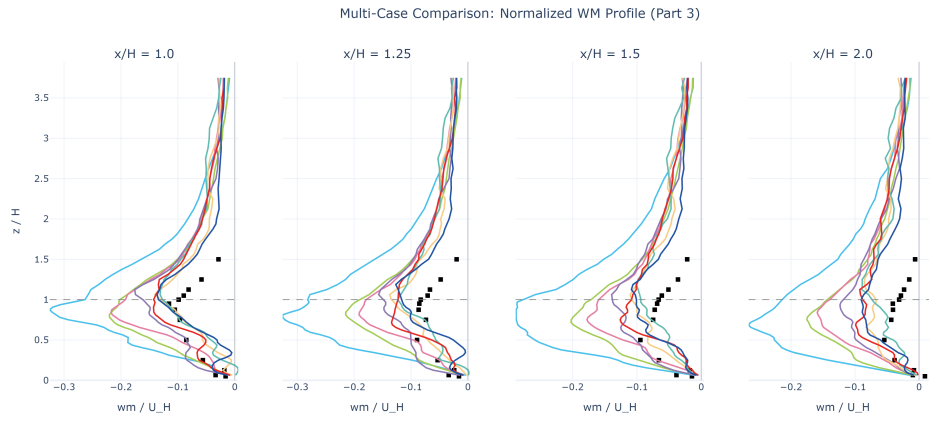
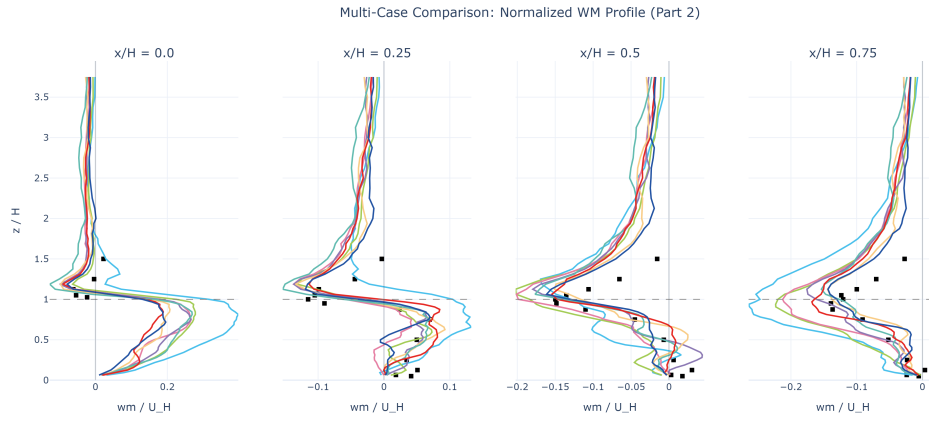
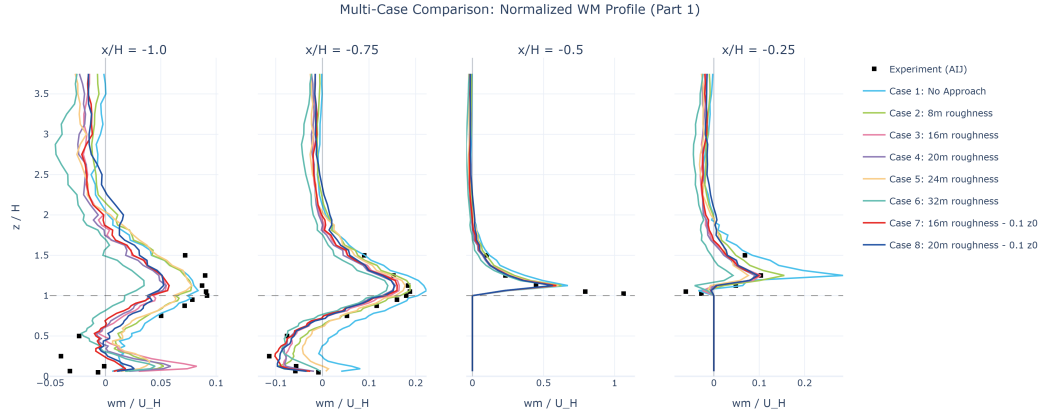


Figure 8: Normalized mean spanwise velocity ($\bar{w}/U_H$) profiles at (a) $x/H = -1.0$ to -0.25 , (b) $x/H = 0.0$ to 0.75 , and (c) $x/H = 1.0$ to 2.0 .

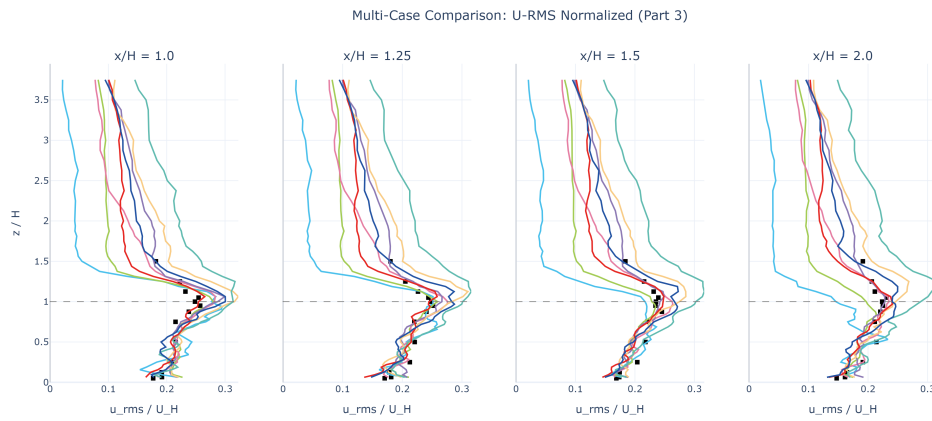
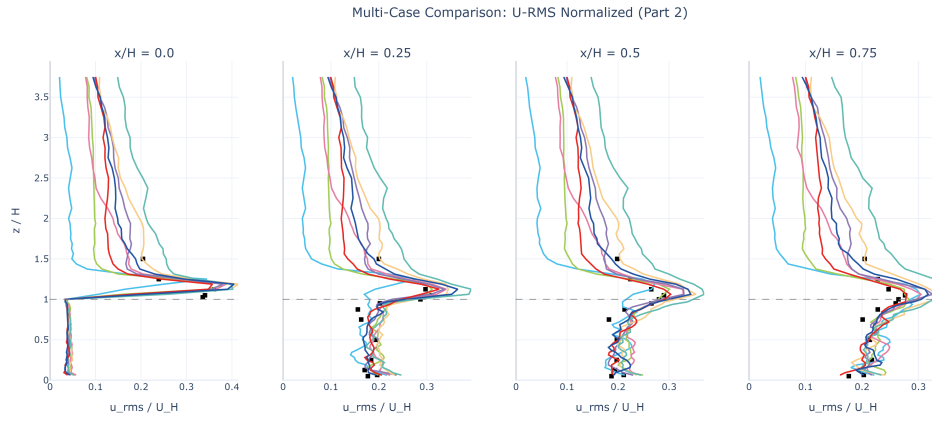
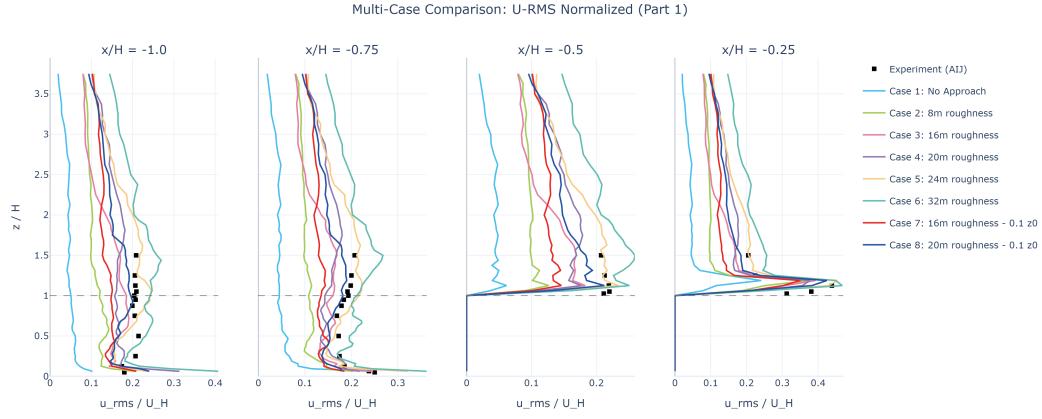


Figure 9: Normalized streamwise velocity fluctuations (u_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).

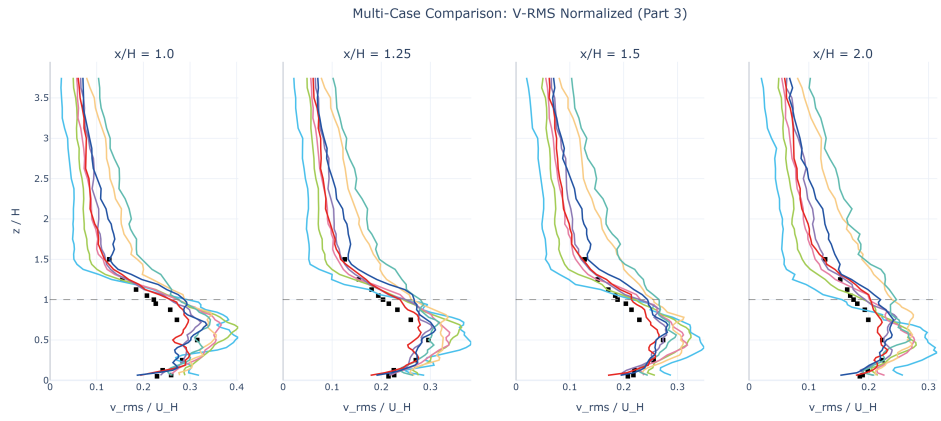
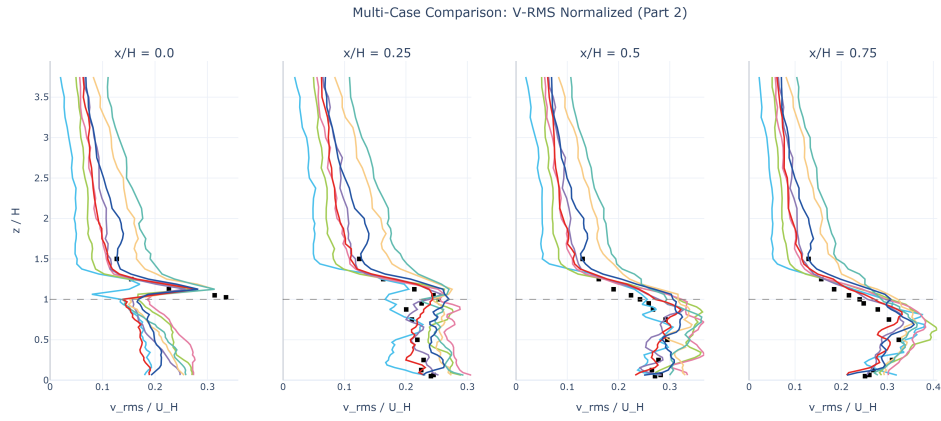
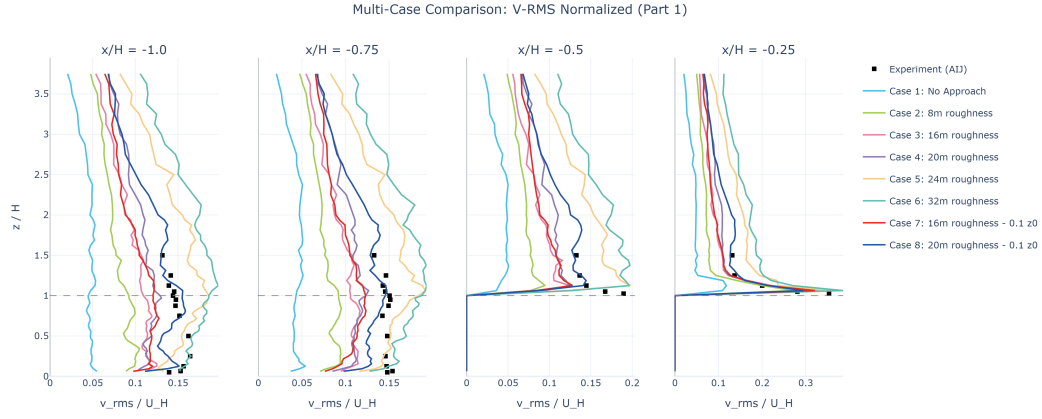


Figure 10: Normalized streamwise velocity fluctuations (v_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).

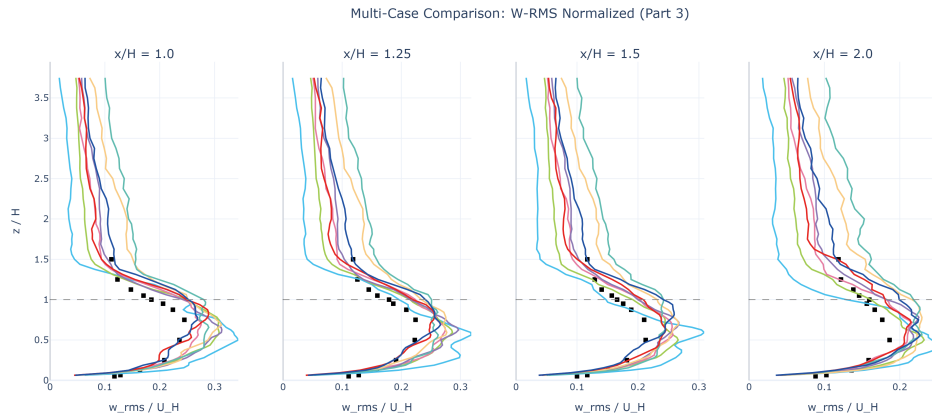
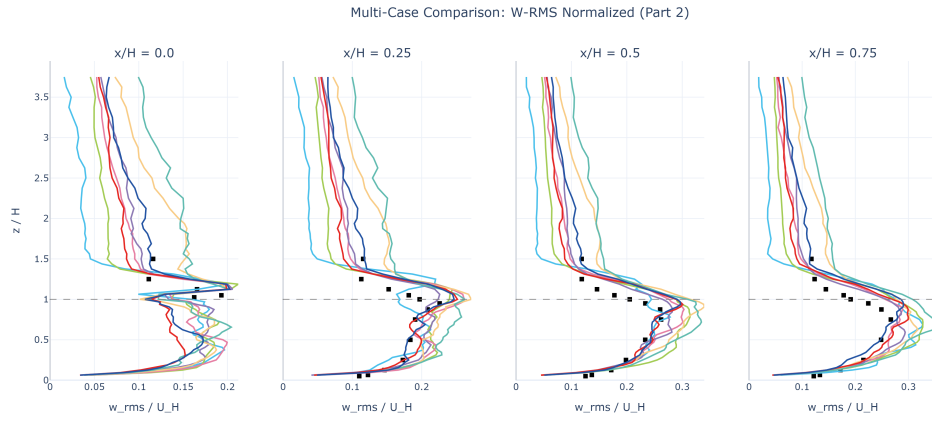
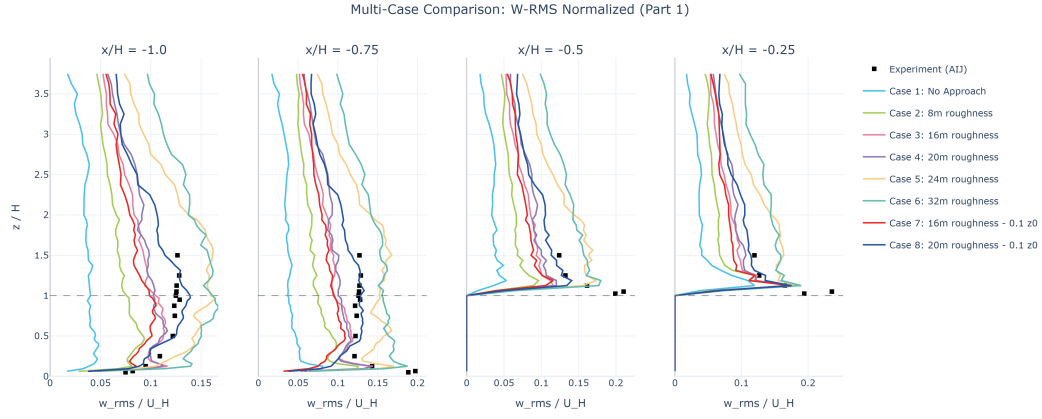
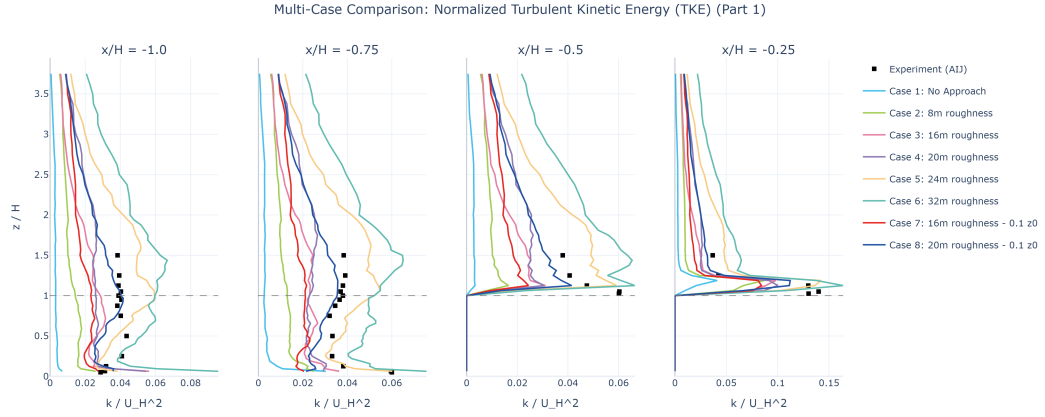
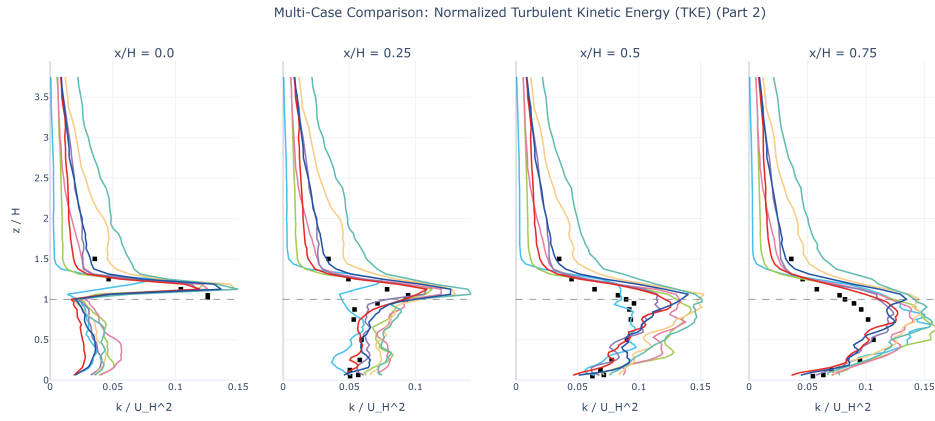


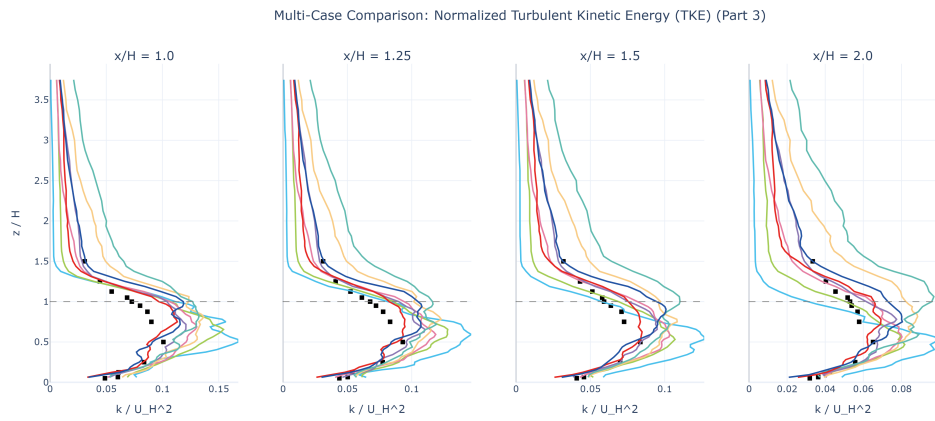
Figure 11: Normalized streamwise velocity fluctuations (w_{rms}/U_H) at (a) Upwind/Rooftop, (b) Leeward Side ($x/H = 0$ to 0.75), and (c) Downwind Wake ($x/H = 1.0$ to 2.0).



(a)



(b)



(c)

Figure 12: Normalized Turbulent Kinetic Energy (k/U_H^2) profiles at (a) Upwind locations ($x/H < 0$), (b) Near Wake locations ($x/H \geq 0$), and (c) Far Wake locations.

Combined Validation: Mean Velocities

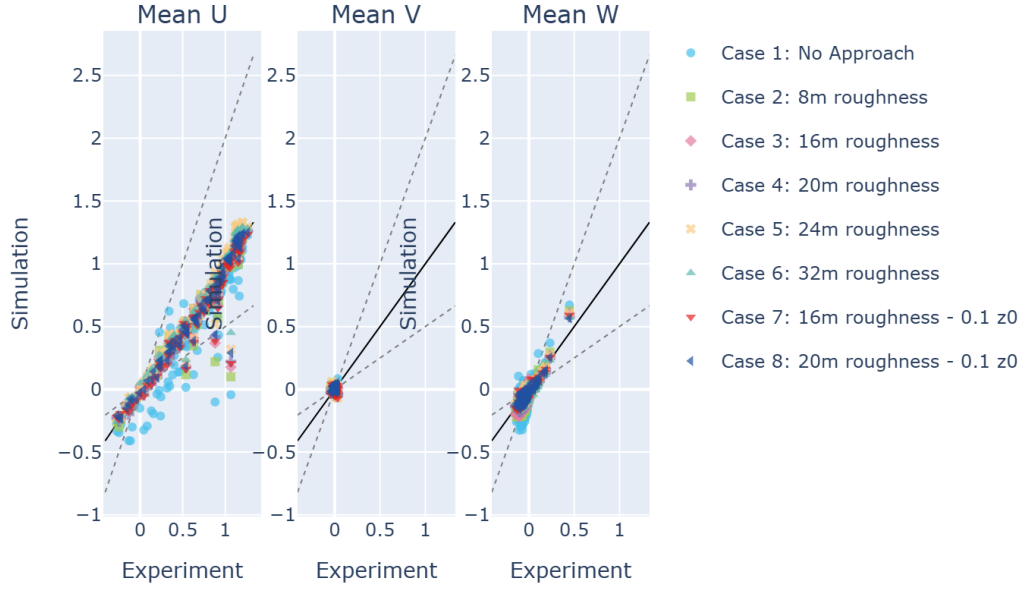


Figure 13: Scatter plot comparison of normalized mean velocity components $(\bar{u}, \bar{v}, \bar{w})$ against AIJ experimental data across all cases. Solid lines represent the 1:1 correlation, while dashed lines indicate the $FAC2$ boundaries.

Table 4: Concentration validation metrics across all approach flow cases. Underlined values indicate performance below the standard thresholds ($q < 0.66$, $FAC2 < 0.50$).

Case	N	q	FAC2	MAE	RMSE
No Approach	119	<u>0.387</u>	<u>0.496</u>	1.4737	3.6180
8m roughness	119	<u>0.403</u>	0.513	1.6328	5.6510
16m roughness	119	<u>0.395</u>	0.588	1.0878	2.7165
20m roughness	119	<u>0.353</u>	0.504	1.3134	3.7468
24m roughness	119	<u>0.387</u>	0.538	1.0368	2.1463
32m roughness	119	<u>0.353</u>	0.529	2.0231	9.4304
16m roughness (z0 0.1)	119	<u>0.420</u>	0.605	1.0131	2.7927
20m roughness (z0 0.1)	119	<u>0.395</u>	0.538	1.1069	2.2813

Combined Validation: RMS Velocities

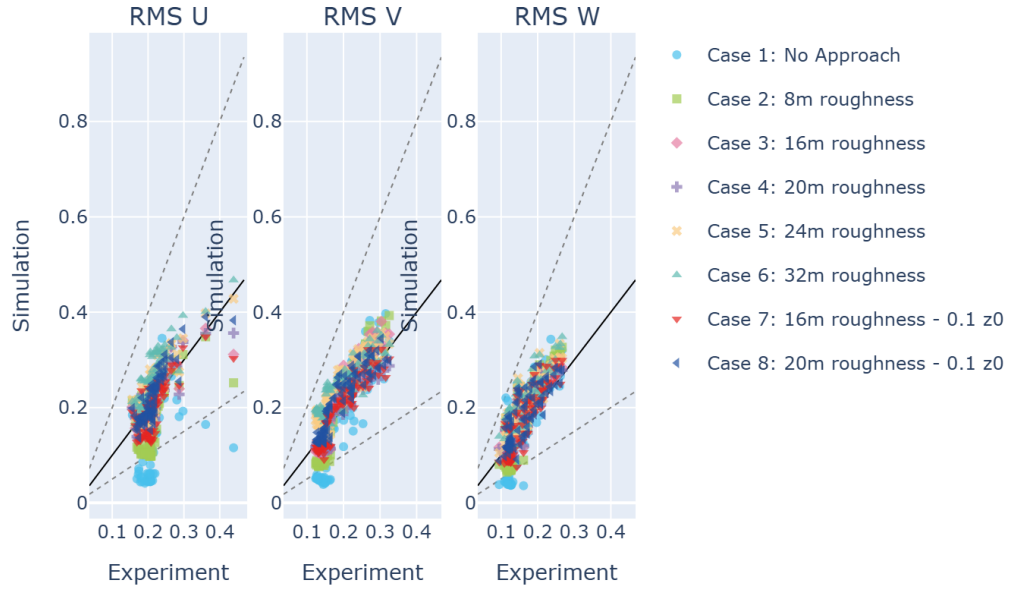


Figure 14: Validation scatter plot for RMS velocity fluctuations ($u_{rms}, v_{rms}, w_{rms}$), demonstrating the solver's ability to capture turbulence intensity across the domain.

Combined Validation: TKE

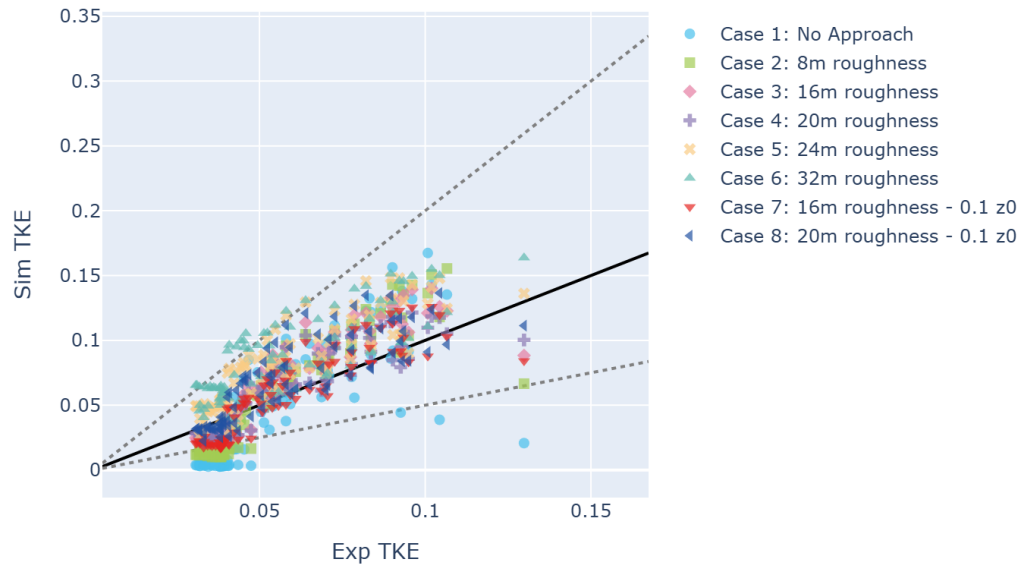


Figure 15: Correlation of predicted vs. observed Turbulent Kinetic Energy (k), highlighting the impact of the refined z_0 and roughness configurations on energy modeling accuracy.

standard, peaking at only 0.420 in Case 7. While the $FAC2$ values mostly meet the 0.50 threshold (with the exception of the "No Approach" baseline), they are considerably lower than the scores achieved for the flow field. These results suggest that the current Lagrangian particle tracker (LPT) within the GPGPU-LBM solver may not yet fully capture the complex dispersion characteristics of pollutants in an urban environment. This discrepancy could be attributed to the simplified physical modeling of generic particles in the solver, whereas the experimental data were collected using gases that exhibit different diffusion behaviors. Furthermore, the high RMSE observed in Case 6 (9.4304) suggests that even with optimized inflow roughness, the modeling of pollutant concentration remains highly sensitive and requires more specialized treatment of gas-phase physics or turbulent mass diffusion to align with physical observations.

Kanda-sensei comment 1: The results on the wind are actually quite good because with the uniform grid in LBM, it should be very difficult to reproduce results since in other LES, the grid becomes very fine as it gets closer to the building. Note this for the discussion in the thesis.

Kanda-sensei comment 2: The results on the concentration may not be very accurate due to the 1. grid issues, 2. the LBM is perfect passive scalar, the experiment uses C_2H_2 and does not specify the particle release momentum or speed. Better to clarify in the discussion and in the presentation, that I am using a perfect passive scalar.

5 Conclusion

Overall,

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